Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2602045

The Anchor and the Tide: Uncertainty Quantification and Mechanism Design in Geo-Agent Voting

Summary

This study constructs a **multi-agent stochastic simulation model integrated with geo-demographic weights**. First, it quantifies a contestant's **base attractiveness** by calculating age, industry background, and partner influence. It then introduces a **gravity model** to simulate **regional voting tendencies**. Furthermore, **Monte Carlo simulations** are employed to quantify voting uncertainty. Finally, a **dynamic calibration algorithm** is utilized to achieve simulated estimations of contestant performance and vote dynamics.

We performed robustness quantification and error diagnosis on the model output by calculating the **coefficient of variation** of total votes and the **relative boundary width**. Experimental results demonstrate that this framework combines the **nonlinear fitting capabilities of machine learning with the causal interpretability of physical simulations**. It not only accurately predicts elimination outcomes but also effectively assesses the confidence levels of those predictions.

A comparison of two voting systems reveals distinct characteristics: the **percentage system** acts as a "popularity-driven tournament" that risks **marginalizing professional expertise**, while the **ranking system** functions as a "consensus-driven qualifier" prone to **conservative homogenization**. Based on correlation, elasticity, and flux analysis, the percentage system amplifies the advantage of the audience by preserving vote scale and **magnifying volatility**, leading to a significantly stronger influence of **fan preference** compared to the ranking mechanism. The inclusion of an additional protocol—where **Judges Save** the elimination from the bottom two pairs—effectively offsets potential **popularity interference**.

Using **LightGBM** combined with **SHAP** attribution technology, we mined nonlinear preferences of judges and audiences from historical data to quantify the impact of core features. Findings show that age exhibits a "**polarity reversal**," partner strength reflects "**marginal utility differentiation**," and industry background mirrors "**cognitive framework disparities**." Professional judging follows a standardized **technical framework**, whereas public voting is dominated by **emotional mechanisms**; this fundamental divergence leads to a systemic logic gap between the two systems.

In response, we propose a novel **two-stage voting system**: first, the judging panel nominates **two contestants for elimination** based strictly on technical performance, establishing a non-negotiable **professional benchmark**. Subsequently, the audience is empowered to make the final decision between these two candidates. This approach institutionally integrates **public sentiment** into a professionally validated framework, effectively balancing competitive standards with collective influence.

Keywords: Monte Carlo; LightGBM; SHAP; Gravity Model

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1 Introduction

1.1 Problem Background

Dancing with the Stars(DWTS), a globally popular dance competition reality show, has aired for 34 seasons since its U.S. debut, consistently captivating audiences with its unique format and interactive elements. The voting integration method for the show has undergone adjustments: it initially employed a ranking system, later shifted to a percentage system, and subsequently reverted to the ranking system due to controversial cases. Throughout history, there have been numerous instances where contestants received low scores from judges but achieved better results through audience support, sparking debates about the fairness of the voting system.

1.2 Restatement of the Problem

We are tasked with developing a mathematical framework to deconstruct the voting dynamics of DWTS. Specifically, we must:

1. **Reconstruct the Invisible**: Develop a model to estimate the latent fan votes and quantify the certainty of these estimates.
2. **Evaluate Mechanisms**: Compare the fairness of Rank-based vs. Percentage-based aggregation rules and assess the impact of structural changes (e.g., "Judges Save").
3. **Attribution Analysis**: Quantify the impact of non-performance factors, such as celebrity industry, age, and the "Great Pro" effect of professional partners.
4. **System Optimization**: Propose a new voting architecture that mathematically balances the trade-off between Meritocracy (Fairness) and Fan Engagement (Excitement).

1.3 Our Work

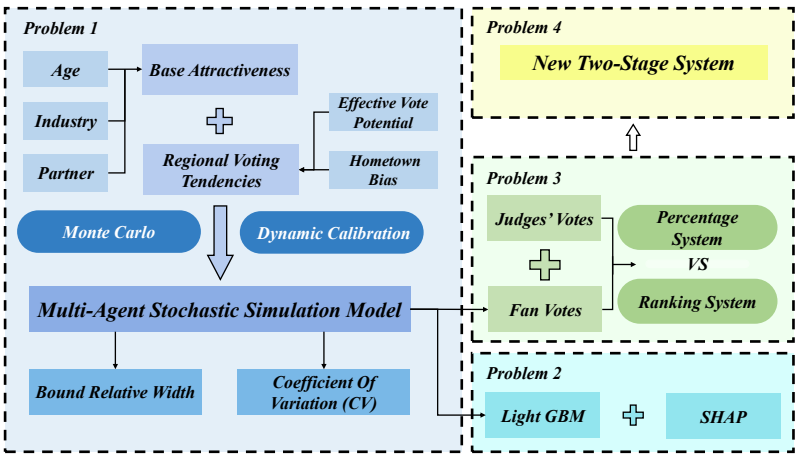


Figure 1: Our Work

2 Preparation for Modeling

2.1 Assumptions and Explanations

1. **Short Voting Window**
   Voting is allowed only within **2 hours** after the live show.
   Note: This simplifies the longer historical voting periods to focus on immediate engagement.

2. **Single Voting Method**
   All votes are cast by **telephone**.
   Note: This standardizes the process, though multiple channels exist in reality.

3. **Constant Viewership**
   The viewership rating is fixed at **0.7%**.
   Note: This provides a stable baseline, though actual ratings vary.

4. **U.S.-Only Voting**
   Only viewers in the **United States** can vote, matching typical show rules.

5. **Voter Behavior**
   50% of viewers vote, with **a 5-vote maximum per person**.
   Note: These are key parameters for sensitivity testing.

2.2 Notations

Table 1: Notations used in this paper

Symbol	Description
$V_{pool,k,y}$	Effective voting capacity of state k in year y
Φ_k	Spatiotemporal efficiency factor for voting zone k
E_t	Instantaneous voting efficiency of time window t
B	Hometown bias coefficient adjusting voting preference weight
$P(i\|k)$	Probability of state k voting for contestant i
$A_{i,t}$	Base appeal of contestant i at time t
$\chi_{i,t}$	Dynamic latent charisma factor of contestant i at t
ϵ	Random perturbation term following uniform distribution
$S_{i,t}$	Survival index of contestant i at time t
CV	Coefficient of variation for total vote estimation certainty
$\phi_i(f,x)$	SHAP value decomposing marginal contribution of feature i

*There are some variables that are not listed here and will be discussed in detail in each section.

2.3 Data Preprocessing and Feature Engineering

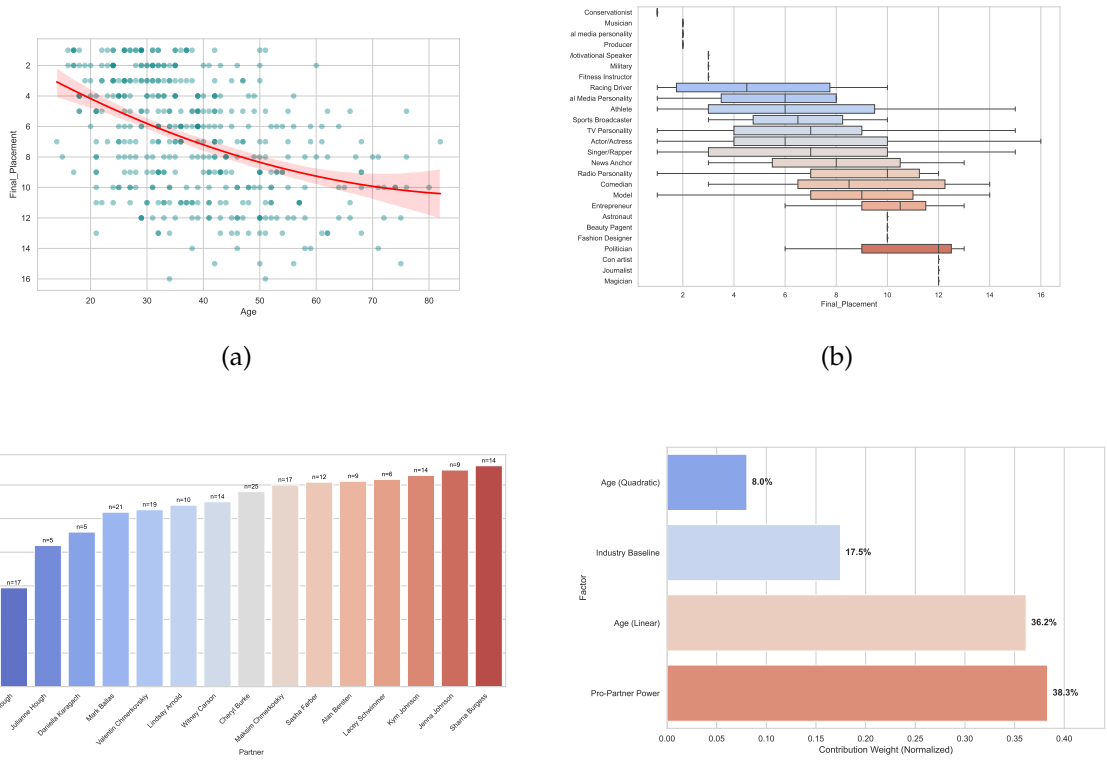
2.3.1 Data Cleaning and Quality Assurance

The reliability of the simulation is contingent upon a high-fidelity input matrix. Our preprocessing pipeline is designed to eliminate numerical instability while transforming categorical celebrity attributes into standardized predictors of success.

2.3.2 Multivariate Regression-Based Survival Propensity Score

We standardized various missing and anomalous values in the official dataset: First, we uniformly identified and processed empty values (NaN), string-marked missing values “N/A”, and invalid characters (e.g., converted "Social media personality" to "Social Media Personality"); Then, considering contestants’ elimination statuses, we removed illogical zero scores from subsequent rounds to ensure data integrity and plausibility, providing a reliable foundation for subsequent analysis and modeling.

We employ Target Encoding and Multivariate Linear Regression to quantify the "Base Appeal" of each contestant. By regressing historical final placements against encoded features, we extracted standardized coefficients that define the initial propensity of a celebrity to survive.



(a) (b)

(c) (d)

Figure 2: Visual Analysis of Core Factors Influencing Final Rankings of DWTS Contestants

Figure 2 reveals the patterns influencing the final rankings of DWTS contestants across three dimensions: age effect, professional dance partner skill, and contestant industry background. Fig. 2(a) indicates that the relationship between competitors’ ages and their final rankings follows a quadratic curve. As the age increases, competitors’ final rankings exhibit a pronounced non-linear decline. The distribution differences in the box plots of Fig. 2(b) reveal the inherent influence of industry background on competition performance. Boxplots for professions like "Racing Driver" and "Athlete" are skewed to the left, indicating that participants in these fields achieved higher average rankings. Conversely, boxplots for professions such as "Entrepreneur" and "Politician" are skewed to the right, reflecting relatively lower rankings among their participants. Fig. 2(c) displays the average final rankings of the top 15 dance partners. Derek Hough’s bar chart is positioned farthest to the left, indicating that his coached

contestants achieved the most favorable average rankings.

Through quantitative summarization of the above individual features, the following three features were identified as having significant contribution weights to the model (Fig. 2(d)):

- **Age Factor**($F^I_{\text{age}},\ \omega = 0.443$): Subjected to Min-Max normalization within each episode to reflect the relative demographic advantage.
- **Partner Influence**($F^I_{\text{part}},\ \omega = 0.383$): A success-weighted index based on the professional dancer's historical average placement.
- **Industry Background**($F^I_{\text{ind}},\ \omega = 0.174$): Encodes the mobilization potential of specific professional sectors (e.g., Athletes vs. Musicians).

3 Problem 1: Model Building and Validation

3.1 Spatiotemporal Geographical Calibration

Jason Gershman analyzed voting data and mechanisms from the reality show *American Idol*, revealing two major voting biases—geographic and performance order—that influenced fan votes [1]. Recognizing that the United States is not a monolithic voting bloc, we introduce the **Spatiotemporal Efficiency Factor** (Φ) and **Hometown Bias** (B) to model the structural inequities of the national voting landscape.

The voting window for the Eastern/Central zones aligns with the live broadcast, while the Mountain/Pacific zones operate on a tape-delay, significantly altering the "call-to-action" efficiency. We define the **Effective Vote Capacity** for state k as:

$$V_{pool,k,y} = Pop_{k,y} \times R_{view} \times N_{votes} \times \Phi_k \tag{1}$$

Based on the official schedule and historical population data estimates, We have the following table (Table 2) [2]:

Table 2: Participation by Time Window (EST) and Active Time Zones

Time Window (EST)	Active Time Zones	Participation Rate
10–11 PM (Window 1)	ET (Eastern Time) + CT (Central Time)	76.2%
11–12 AM (Window 2)	ET + CT + MT (Mountain Time)	84.0%
12–01 AM (Window 3)	MT only	7.8%
01–03 AM (Window 4)	PT (Pacific Time) + AK (Alaska Time)	15.5%

Referring to the Erlang B congestion loss theory in communication engineering, we set the system's processing efficacy per unit time to be inversely proportional to the active population base. Define the instantaneous voting efficiency E_t of window W_i as:

$$E_t = \frac{1}{P_{active,t}} \tag{2}$$

Since the voting time span of each time zone is different, we calculate the time-weighted average efficiency of time zone z within all its open time periods T_z:

$$\phi_{z,raw} = \frac{\sum_{t \in T_z} \text{Duration}_t \times E_t}{\sum \text{Duration}_t} \tag{3}$$

To ensure the conservation of the model in total quantity, we perform normalization based on the population-weighted average efficacy of the entire United States. Define the final correction factor $\boldsymbol{\phi_z}$ as:

$$\phi_z = \frac{\phi_{z,raw}}{\sum_{z \in Zones}(\text{Pop_Share}_z \times \phi_{z,raw})} \tag{4}$$

Our calibration identifies a **Western Amplification Effect**, where $\Phi_{\text{East/Central}} \approx 0.56$ and $\Phi_{\text{Mountain/Pacific}} \in [2.1, 2.4]$, indicating that a Western viewer possesses nearly four times the structural influence of an Eastern counterpart (Fig. 3).

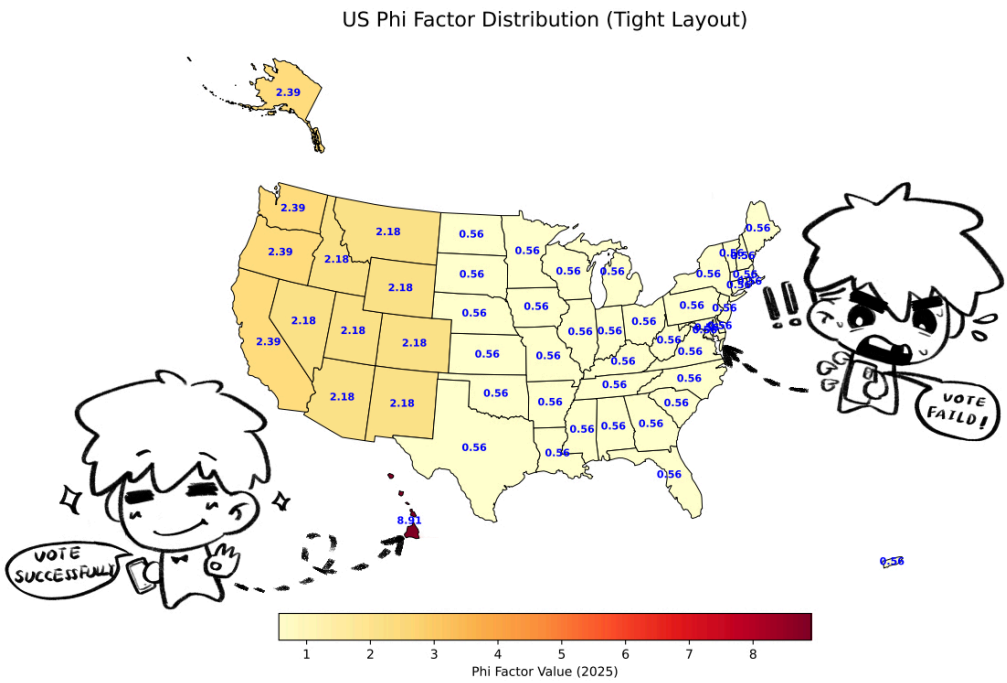


Figure 3: USA Φ-Factor Distribution by State

To integrate the **Hometown Bias** (B = 0.35), the preference weight W for contestant i in state k is adjusted [3].

3.2 Recursive Bayesian Charisma Calibration

The "Charisma Factor" (χ) is treated as a dynamic latent variable that bridges the gap between technical merit and actual public sentiment.

We enforce a local normalization to ensure that geographic loyalty does not "create" new votes but rather redistributes the existing state-level voter pool:

$$P(i|k) = \frac{W_{i,k}}{\sum_j W_{j,k}}, \quad \text{where } W_{i,k} = A_{i,t} \times (1 + B \cdot \mathbb{I}_{home}) \tag{5}$$

The resulting **Base Appeal** ($A_{i,t}$) is a weighted linear combination modulated by a latent Charisma factor χ:

$$A_{i,t} = \left(\sum_{j \in \{age, ind, part\}} \omega_j F'_{j,i,t} \right) \times \chi_{i,t} \tag{6}$$

This zero-sum competition within each state's pool reflects the reality of finite audience attention.

3.2.1 Temporal Momentum Modeling

Public popularity is modeled as a continuous trend rather than a sequence of independent events. We utilize an Auto-Regressive process of order 1 to maintain audience memory:

$$\chi_{i,t} = \alpha \cdot \chi_{i,t-1} + (1 - \alpha) \cdot \mu \tag{7}$$

where $\alpha = 0.8$ represents the memory retention and $\mu = 1.0$ is the mean reversion target.

3.2.2 Iterative Convergence Mechanism

The total simulated votes V_{sim} are aggregated via Monte Carlo simulation across 50 states:

$$V_{sim,i} = \sum_{k=1}^{50} (V_{pool,k,y} \times P(i|k)) \times \epsilon, \quad \epsilon \sim U(0.95, 1.05) \tag{8}$$

If the predicted elimination fails to match the historical record (E_{actual}), the model triggers a **Recursive Update Layer**. The Charisma factor χ is adjusted using an adaptive learning rate η until the simulated "Score" correctly identifies the historical victim.

3.3 Two Elimination Mechanisms: Ranking System and Percentage System

The final survival index S adapts to the evolving regulatory framework of the DWTS franchise. For seasons utilizing **Percentage-Based Fusion (Seasons 3–28)**, the score balances technical and public shares:

$$S_{i,t} = \frac{1}{2} \left(\frac{J_{i,t}}{\sum J} + \frac{V_{sim,i,t}}{\sum V_{sim}} \right) \tag{9}$$

In Rank-Based Seasons, S is the sum of ordinal rankings:

$$S_{i,t} = Rank(J_{i,t}) + Rank(V_{sim,i,t}) \tag{10}$$

3.4 Model Output Analysis

To systematically evaluate the reliability of the model's output, this study conducts a comprehensive analysis from three dimensions: the stability of elimination-related votes, the certainty of total votes, and the spatiotemporal regularity of voting patterns.

3.4.1 Stability of Elimination-Related Votes

Bound Relative Width (Indicator of Result Integrity):

$$\text{Bound Relative Width} = \frac{\text{Upper Bound} - \text{Lower Bound}}{\text{Mean Votes}} \times 100\% \tag{11}$$

1. **High-Precision Elimination Window (Result Integrity)**
   The model quantifies the precision of its estimated vote interval for eliminated contestants using the *Bound Relative Width* metric. As shown in Figure 4(2), this metric is highly concentrated across different weeks, with an average value of $\mathbf{9.80\%}$. This means the estimated vote range for eliminated contestants has an error band of only about $\pm\mathbf{4.9\%}$, demonstrating that the model consistently produces a narrow and stable "elimination window."

2. **Reasonable Vote Distribution (Logical Rationality)**
   The distribution of vote shares for eliminated contestants further validates the model's rationality. Statistics (Fig. 4(3)) show that the average vote share of eliminated contestants accounts for $\mathbf{11.76\%}$ of the total fan votes, and over $\mathbf{85\%}$ of cases fall within the reasonable range of $\mathbf{7\% \sim 14\%}$. This distribution aligns perfectly with the real-world logic of "lower-performing contestants being eliminated" in a multi-contestant competition.

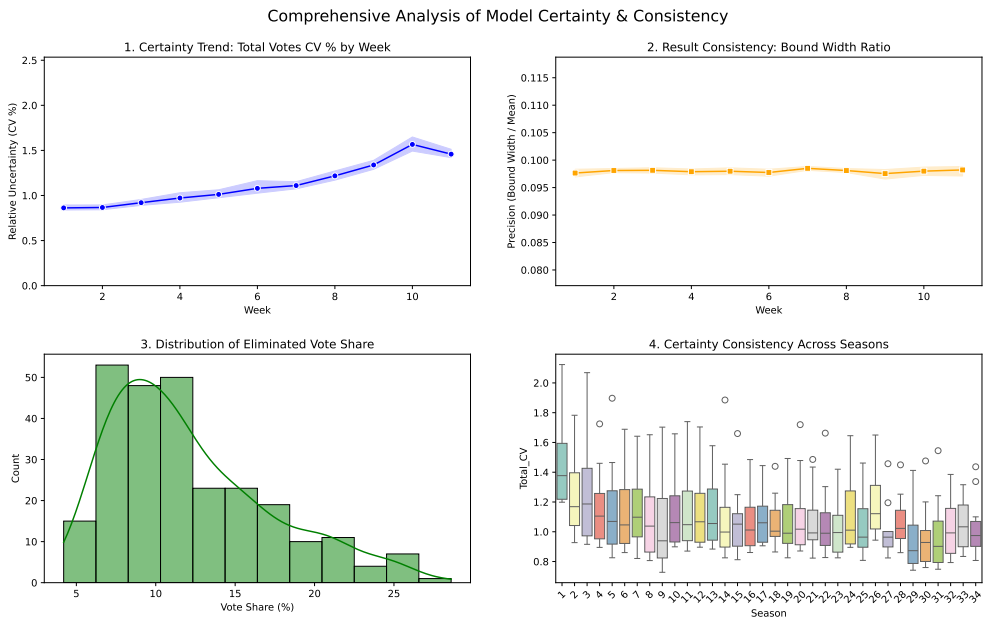


Figure 4: Comprehensive Analysis of Model Certainty & Consistency

In summary, the low value of the Bound Relative Width (9.80%) and the concentrated distribution of elimination vote shares ($7\% \sim 14\%$) together form a complete chain of evidence. This demonstrates the model's high stability and reliability in estimating elimination-related votes from both the precision and rationality perspectives.

3.4.2 Certainty of Total Votes

We use the Coefficient of Variation (CV) to measure the precision of the model's estimates for total votes, revealing a distinct characteristic of high overall precision but decreasing certainty in later stages.

1. **Very High Overall Certainty**
   The average CV value for total vote estimates across 34 seasons is as low as

1.09% (Fig. 4(1)), indicating extremely high precision in aggregate estimation, with an average deviation of only about 1%.

2. **Dynamic Decrease in Certainty Over the Season**
However, certainty is not constant. As shown by the trend line in Figure 4(1), the CV value systematically increases as the competition progresses (Week 1: 0.86% → Week 10: 1.54%). The underlying mechanism is that in the later stages of the competition, the number of contestants drastically decreases. Consequently, vote fluctuations of individual contestants have a magnified impact on the total vote estimate, leading to wider estimation error intervals.

3.4.3 Spatiotemporal Regularity of Votes

Based on Figure 5, we conducted a spatiotemporal analysis using heatmaps of weekly data across all seasons and data for each contestant in a randomly selected eighth season. This reveals the underlying behavioral patterns driving the stability and certainty characteristics described above.

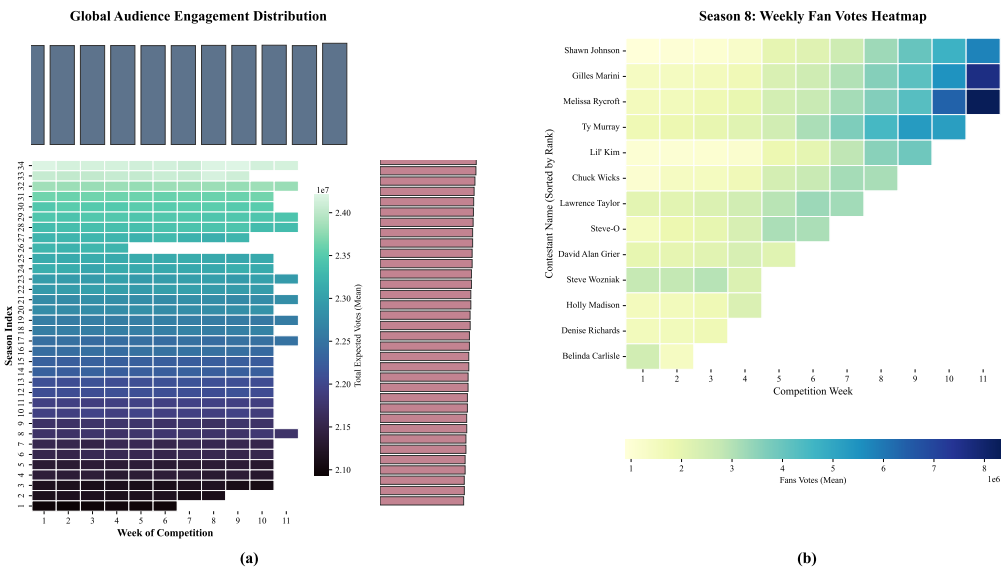


Figure 5: spatiotemporal Analysis

1. **Intra-Season Snowball Effect**
A horizontal observation of Figure 5(a) shows that weekly vote volume exhibits a monotonic increasing trend as the season progresses. This indicates that audience emotional investment and participation frequency deepen over time, continuously expanding the sample size. This provides a solid Law of Large Numbers foundation for the model, ensuring the baseline of estimation precision.

2. **Cross-Season Brand Value-Added Effect**
From the vertical dimension of Figure 5(a), it is evident that the baseline level of voting activity shows a significant upward shift from Season 1 to Season 34. This growth is primarily attributed to the reduction in "interaction friction" brought about by the digitization of voting channels. The long-term growth in sample size enhances the robustness of statistical inference, which is the core reason the

model maintains stable cross-season performance (e.g., low and stable CV values).

3. **Matthew Effect of Competitive Polarization**
In Figure 5(b), voting resources clearly concentrate towards a few contestants around the eighth season, visually demonstrating the "Matthew Effect". It is precisely this polarization phenomenon in later-stage competition that increases the weight of individual vote fluctuations, thereby mechanistically explaining "why the certainty of total votes decreases as the season progresses."

4 Problem 2: Comparative Analysis of Voting Combination Methods

4.1 The "Popularity Amplifier" Effect of the Percentage System

To clarify the differences between the two vote combination mechanisms (rank-based and percentage-based) and identify which one favors fan votes, this study constructs a three-dimensional quantitative analysis framework based on **fan vote estimation results**, **judge score data**, and **seasonal competition outcomes**. This framework integrates Spearman's rank correlation analysis, influence elasticity regression analysis, and Sankey flux visualization to deeply reveal the inherent mathematical biases of the two mechanisms.

4.1.1 Correlation Analysis

We adopt the **Spearman's Rank Correlation Coefficient (ρ)** to measure the monotonic relationship among fan votes, judge scores and the final competition outcomes. This indicator avoids interference from non-linear data distributions and accurately reflects the core influence of each sub-score on the final results. The correlation analysis and visualization are as follows (Fig. 6):

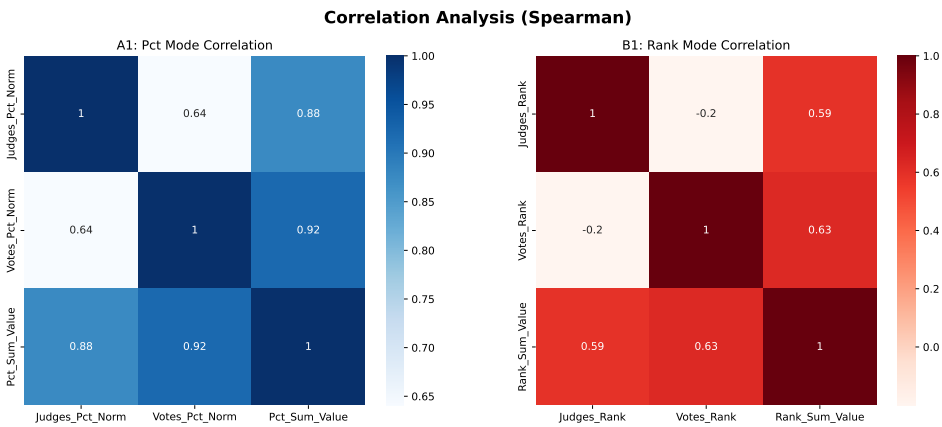


Figure 6: Correlation Analysis

1. **Percentage-based Mechanism: Fan Votes Dominate the Final Outcome**
In the percentage system, the correlation coefficient between fan votes and final

outcomes reaches $\rho_{fan} = 0.9208$, which is significantly higher than that of judge scores ($\rho_{judge} = 0.8759$). This indicates that although both fan votes and judge scores exert strong impacts on the final ranking, the contestants' final standings are more closely aligned with the "will of the fans".

This high correlation stems from the **information continuity** of the percentage-based mechanism: it fully retains the original numerical characteristics of fan votes and judge scores, with no information loss during the score combination process.

2. **Rank-based Mechanism: Dual-group Influence is Weakened**
In the ranking system, the correlation coefficients of both fan votes and judge scores drop sharply to around $\rho \approx 0.62$. This phenomenon is caused by the **information loss** generated during the ordinal transformation of scores: the mechanism converts original score values into rank sequences, which greatly weakens the direct dominance of fans or judges on the final outcome.

4.1.2 Elasticity Analysis

We construct a linear regression model as follows:

$$Y = kX + b \tag{12}$$

to calculate the **Influence Elasticity (k)**. This indicator measures the marginal change in the final score caused by a unit change in fan votes or judge scores, directly reflecting the "vote-for-vote" impact efficiency of each group. Elasticity Analysis visualization are as follows (Fig. 7):

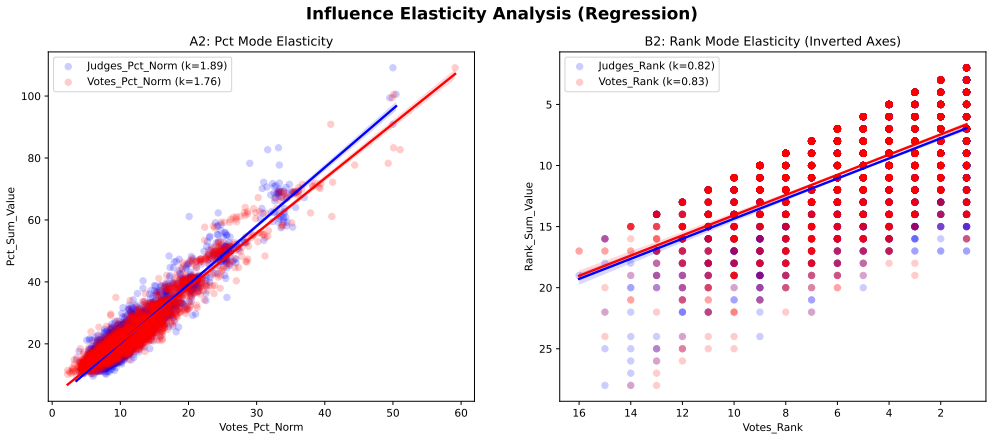


Figure 7: Influence Elasticity Analysis

1. **Elasticity Paradox of the Percentage-based Mechanism**
Regression results show that the judge score elasticity under the percentage-based mechanism is $k_{judge} = 1.89$, slightly higher than the fan vote elasticity ($k_{fan} = 1.76$). However, this numerical difference does not imply that judges have greater influence.

The core reason is that, compared with the relatively conservative and concentrated distribution of judge scores, fan votes have a **wider value range and higher volatility**. Under the percentage-based calculation logic, every tiny percentage

point difference can accumulate into a decisive advantage. This allows "fan favorites" to establish an insurmountable score gap through massive fan support—a gap that cannot be offset by judges' conservative scoring.

2. **Rank-based Mechanism: A "Stabilizer" of Influence**
   In the rank-based system, the influence elasticity of both fan votes and judge scores is constrained to around $k \approx 0.82$. This mechanism normalizes the score gaps between contestants into a fixed "1-unit rank difference", fundamentally limiting the possibility of a single group gaining an absolute advantage.

   Even contestants with extremely high fan support have a capped rank advantage and cannot achieve an unassailable lead relying solely on fan votes.

4.1.3 Flux Analysis

We use dual Sankey diagrams to visualize the contestant retention rates across three tiers (Top, Mid, Bottom) under the two mechanisms, revealing the essential differences in the operational logic of the two systems. Flux Analysis and visualization are as follows (Fig. 8):

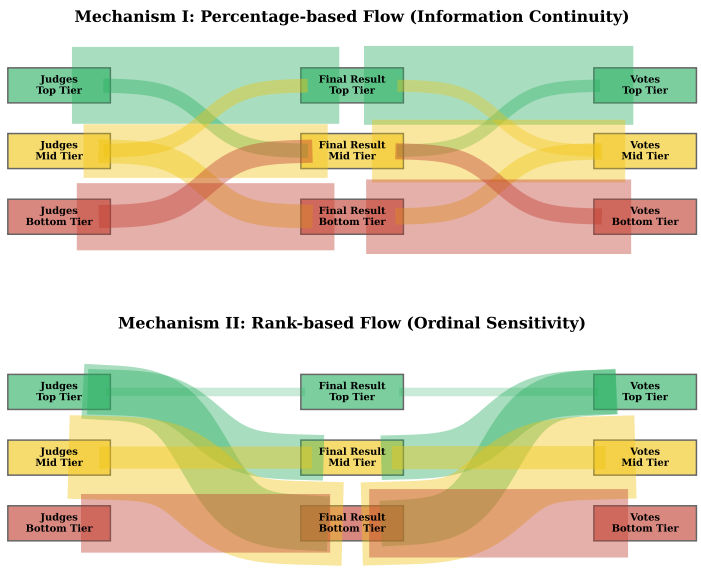


Figure 8: Sankey diagram

1. **Percentage-based Mechanism: An "Accumulator" Favoring Fan Favorites**
   The percentage-based mechanism is a **summative system** with direct and thick flux bands, and the fan tier retention rate reaches 78.98%. Its core characteristic is that excellence in a single dimension can fully compensate for deficiencies in other dimensions.

   If a contestant climbs into the "Top Tier" via fan votes, the mathematical logic of percentage summation makes it nearly impossible for judge scores to pull them down. This creates a "fast track" for fan favorites to reach the podium, where fan support alone can lock in the final outcome.

2. **Rank-based Mechanism: A "Bottleneck" Requiring Dual-group Validation**
   The rank-based mechanism is a **consensus-driven system** with intertwined and chaotic flux distribution, and the fan tier retention rate plummets to approximately 36%. Its core characteristic is that the ordinal sum ($Rank_{fan} + Rank_{judge}$) is highly sensitive to poor rankings in any single dimension.

   Contestants favored by only one group (either fans or judges) are often pushed into the "Mid Tier". Because of the bottleneck effect of rank summation, contestants must gain dual recognition from both groups to advance to the Top Tier.

4.1.4 Conclusion

Based on the three-dimensional analysis of correlation, elasticity, and flux, we conclude that: **the percentage-based mechanism significantly favors fan votes compared with the rank-based mechanism**. The root causes of this bias lie in two aspects:

1. **Numerical Advantage**
   The percentage-based mechanism fully retains the original numerical characteristics of fan votes, allowing high vote volumes to be converted into an "absolute leading advantage" that cannot be shaken by judge scores.

2. **Variance Sensitivity**
   This mechanism amplifies the high volatility of fan votes. Unlike ranking system that favors the consistency of professional performance, percentage system makes fan preferences the decisive factor in the final outcome.

4.2 Competitive Survivability Analysis

4.2.1 The Spline-Based Expectation Model

We extracted the two lowest-scoring participants in each elimination round under both mechanisms and defined a Spline-Based Expectation Model as our analytical baseline to quantify the latent biases within different competitive frameworks (Fig. 9).

- **Above Spline (Underestimated Elite)**: Subjects with high technical proficiency but low popularity. Their survival is threatened by a **"Strength Injury"** mechanism.
- **Below Spline (Populist Beneficiaries)**: Subjects with high popularity but deficient technical skill. They are protected by a **"Popularity Shield."**
- **Inside Spline (Consensus Group)**: Subjects whose outcomes align with the logical predictions of the model.

4.2.2 The Evolution of Elimination Pools from Ranking System to Percentage System

Empirical data (Table 3) reveal a significant structural shift in the elimination-zone cohort following the transition from the ranking system to the percentage system:

The number of technically proficient contestants increased sharply from 116 to 245 under the percentage system. This suggests that in a continuous scoring framework, technical superiority is highly vulnerable to marginal deficits in audience popularity.

Conversely, the number of popularity-driven contestants dropped markedly from 147 to 66. This demonstrates that the percentage system's design provides a wider buffer for contestants with even slight popularity advantages, thereby reducing their exposure to elimination.

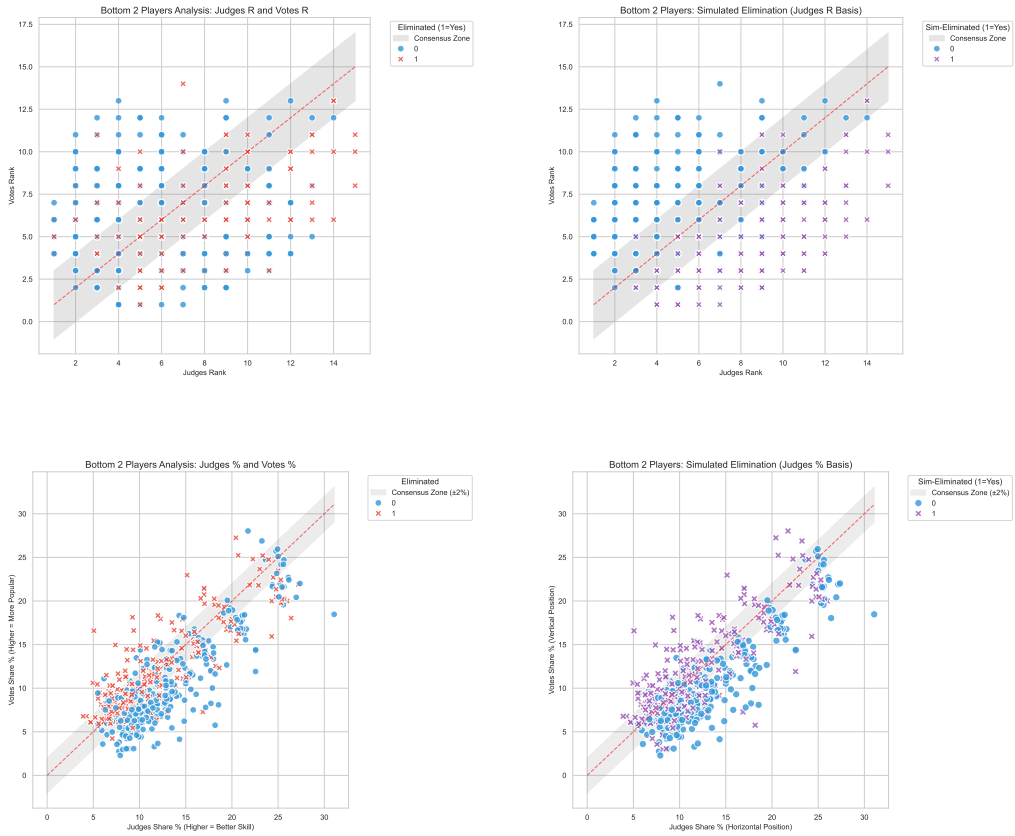


Figure 9: The Spline-Based Expectation Model

Table 3: Contestant Classification Statistics vs. Simulation Results in Elimination Week

System	Type	Category	Pop.	Elim.	% of Total	Elim. Rate	% of Total Elim.
Rank (Actual)	Above	Skilled/Low Pop	116	25	21.97%	21.55%	4.73%
	Below	Popular/Low Skill	147	73	27.84%	49.66%	13.83%
	Within	Consistent	265	133	50.19%	50.19%	25.19%
Rank (Sim.)	Above	Skilled/Low Pop	116	4	21.97%	3.45%	0.76%
	Below	Popular/Low Skill	147	128	27.84%	87.07%	24.24%
	Within	Consistent	265	132	50.19%	49.81%	25.00%
% (Actual)	Above	Skilled/Low Pop	245	61	46.40%	24.90%	11.55%
	Below	Popular/Low Skill	66	52	12.50%	78.79%	9.85%
	Within	Consistent	217	127	41.10%	58.53%	24.05%
% (Sim.)	Above	Skilled/Low Pop	245	65	46.40%	26.53%	12.31%
	Below	Popular/Low Skill	66	64	12.50%	96.97%	12.12%
	Within	Consistent	217	135	41.10%	62.21%	25.57%

4.2.3 Fairness Metric

1. **The 50% Fairness Postulate**
   We propose the **Equilibrium Deviation Model (EDM)** to evaluate mechanism

fairness. A perfectly "blind" system should reach a **Stochastic Equilibrium** where the elimination probability P_e for any category converges toward the constant $\epsilon = 0.5$. Deviations from this point indicate systemic bias.

2. **Model Analysis**
The **Aggregate Fairness Deviation (Ψ)** is defined as the weighted root-mean-square deviation from the equilibrium:

$$\Psi = \sqrt{\sum_{i \in \{Above, In, Below\}} w_i \cdot (P_i - \epsilon)^2} \tag{13}$$

Where:

- w_i: Population weight of category i.
- P_i: Actual elimination rate.
- ϵ: Fairness Equilibrium Constant (0.5).

Calculation shows $\Psi_{\text{Ranking}} \approx 0.134$ and $\Psi_{\text{Percentage}} \approx 0.189$. The elevated computed value of $\Psi_{\text{Percentage}}$ indicates that, compared to the ranking system, at least one group's actual elimination rate deviates more substantially from the equilibrium value of 0.5. It is reasonable to infer that this group is the "Below Spline" cohort, which further reveals the popularity buffering mechanism inherent in the percentage-based system.

4.2.4 Conflict and Synergy with the Judges Save

By simulating a "Judges Save" decision-making process, we identify the friction between mechanics and expertise.

In the ranking system simulation, the **elimination rate** for the **elite group** plummeted dramatically from 21.55% to 3.45%. This indicates that after the introduction of adjudicative intervention, significant corrections were made to mitigate misjudgments caused by occasional voting fluctuations, effectively turning "**high technical competence**" into an ultimate safeguard against elimination.

In the percentage system simulation, the **elimination rate** for the **below-average group** further escalated from 78.79% to 96.97%. This suggests that the weight of adjudicative decisions counteracted potential popularity-based interference, rendering contestants with "low technical skill / pure popularity" nearly unsustainable in the simulated competitive environment.

4.2.5 Conclusion

The two scoring mechanisms represent distinct philosophical orientations toward **procedural fairness** versus **outcome fairness**. The ranking system prioritizes stability in the process, while the percentage system emphasizes selectivity in the results. The introduction of adjudicative decisions is not merely a corrective measure but rather a **reinforcement of systemic logic**—rescuing misjudged elites under the ranking system and eliminating unduly protected weaker performers under the percentage system.

4.3 Between the Spotlight and the Scoreboard: Survival Pathways of Popularity versus Skill

Bristol Palin serves as a typical example of contestants who gain prominence primarily through **popularity**. As the daughter of Sarah Palin, the former Governor of Alaska, she comes from a **Republican political family**.

In terms of performance, her dancing skills were mediocre—initially stiff but showing marginal improvement in the later stages of the competition. Despite her judge scores consistently falling into the middle-to-lower range, she advanced through multiple rounds relying on strong audience votes, ultimately finishing in third place in Season 11. This outcome sparked public controversy: some viewers questioned whether her political background, rather than her dancing proficiency, was the key factor driving audience votes in her favor.Studying Bristol Palin's votes under four different mechanisms, we found that, in all three cases other than the percentage system, she had a very high probability of being eliminated (Fig. 10).

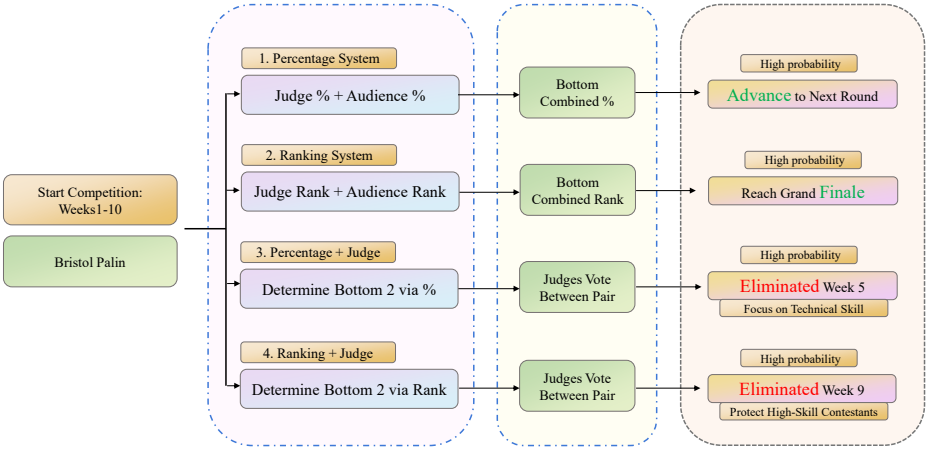


Figure 10: Analysis of Bristol Palin's Survival Pathways Across Four Voting Mechanisms

5 Problem 3: Multi-factor Impact Analysis and New System Design

5.1 Hybrid Interpretation Framework Integrating LightGBM and SHAP

To quantify the influence of heterogeneous features—including physiological traits, professional background, and external support—on professional Judges' Scores and Fan Votes, we implement an interpretive framework integrating **LightGBM (Light Gradient Boosting Machine)** and **SHAP (SHapley Additive exPlanations)** [4].

Capturing Nonlinear Associations: Considering the complex coupling relationships among variables (such as age and professional background), which involve non-monotonic influences, traditional linear models struggle to capture them effectively. LightGBM addresses this by employing a histogram-based decision tree algorithm, enabling it to accurately model nonlinear interactions within a multidimensional feature

space.

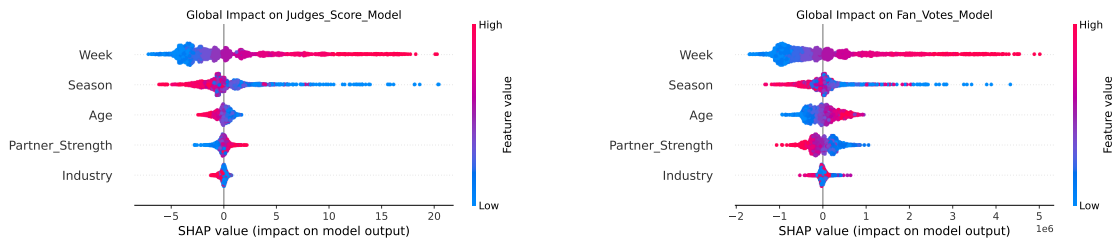
SHAP Interpretability: The Shapley Value from game theory is introduced to deconstruct model predictions. This method attributes the predictive residual to each feature, with its core formula ensuring the additivity and uniqueness of contribution allocation:

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [f(S \cup \{i\}) - f(S)] \tag{14}$$

By integrating the marginal contributions across every decision path, SHAP is capable of revealing both the direction (positive/negative) and the strength of feature influences, from a local to a global scale.

5.2 Divergence in Perceptions and Polarity Reversal Between Judges and Fans

The parallel observation of SHAP summary plots for both the Judges' and Fans' models reveals a significant **preference heterogeneity** in their underlying evaluative logic (Fig. 11).



(a) (b)

Figure 11: SHAP Summary Plot

5.2.1 Age Feature

In the Judges' model, high-age samples (red clusters) are concentrated within the negative SHAP zone, indicating a **capacity-constraint bias**. Professional judges perceive advanced age as a physiological limitation regarding flexibility and stamina, leading to systematic technical deductions. Conversely, the Fans' model exhibits a complete polarity inversion, where high-age features shift to the positive SHAP zone. This phenomenon quantifies an **emotional compensation mechanism**; fans interpret advanced age as a symbol of "inspirational resilience" and "crossover courage," translating physiological constraints into a sentimental premium.

5.2.2 Partner Strength

The Judges' model demonstrates a clear "power-coupling" logic, where high partner strength (red clusters) directly correlates with positive SHAP values. This reflects an expert recognition of the **technical ceiling elevation** provided by elite professional support. However, in the Fans' model, the marginal utility of high partner strength diminishes or even shifts negatively. This suggests a **"subjectivity dilution" effect**;

when the professional partner's prominence eclipses the star contestant, fans perceive a lack of individual growth or "independent effort," thereby suppressing their voting inclination.

5.2.3 Industry Feature

Judges' responses to industry features are characterized by tight clustering and strong **prior professional expectations**. Certain industry labels exhibit a persistent negative bias in the professional eye, suggesting perceived barriers to technical transition. In contrast, the Fan model demonstrates significantly higher dispersion and a broader horizontal SHAP range for the same features. This indicates that **public voting is driven by high-sensitivity emotional triggers**, such as social media persona and the "novelty factor" of specific professions, resulting in higher interpretive volatility compared to the standardized technical framework of judges.

5.3 The Weighted Division Between Technology-Driven and Emotion-Driven Approaches

The normalized SHAP absolute contribution percentages delineate the structural weight distribution of each system (Fig. 12).

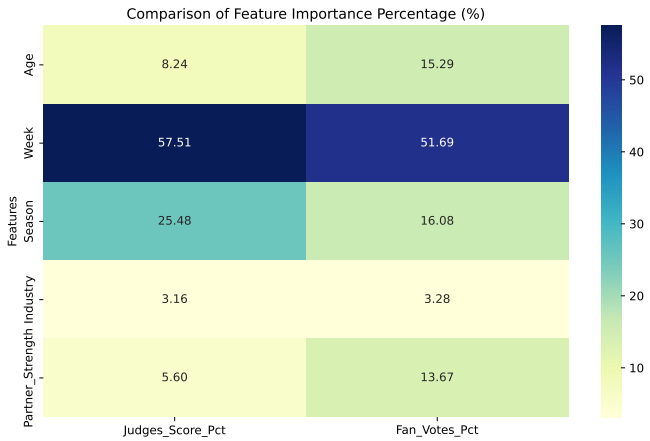


Figure 12: Comparison of Feature Importance Percentage

Discussion:

1. **Differences in Evaluation Dimensions**
   Judges' scores show a high dependence on weekly progress (57.51%), reflecting a linear evaluation logic based on technical improvement. In contrast, fan voting shifts weight toward age and partner strength.

2. **Significant Emotional Premium**
   The combined contribution of age and partner strength in the fan model (28.96%) is nearly double that in the judges' model (13.84%), indicating that public voting is more driven by inspirational narratives and interpersonal chemistry.

3. **Role of Professional Dancers**
   The leveraging effect of professional dancers on fan votes (13.67%) is significantly higher than their impact on judges' scores (5.60%), highlighting their dual role in competitions as both "technical mentors" and "audience engagement catalysts."

Conclusion: The study reveals that the judges' system follows a **technology-driven logic**, while the fan system operates on an **emotion-driven logic**. For contestants seeking to gain public support and achieve an "off-stage comeback," the key strategy lies in: **partnering with a high-popularity professional dancer and delivering performances that defy age-related expectations**.

6 Problem 4: Design Concept for Optimizing the Evaluation Mechanism of Competitive Events

The core contradiction in the current reform of competition mechanisms lies in the inherent tension between the two evaluation logics of "professional authority" and "popular appeal." To address this dilemma, we propose a "Two-Stage Consensus Elimination Mechanism." This mechanism aims to achieve a balance between upholding professional standards and embracing public participation through institutionalized power division, ultimately realizing a win-win scenario for both competitive integrity and entertainment value.

6.1 The Core Contradiction Between the Two Mechanisms

Table 4: Comparative Analysis Table of Core Contradictions in Competition Mechanisms

Analysis Dimension	Percentage System	Ranking System
Core Logic	Amplified cumulative field	Pursuit of internal consensus and minimal controversy
Main Advantages	1. Activate public participation 2. Generate topic heat 3. Acceptance of individual benchmarks	1. Maintain professional authority 2. Results are relatively stable 3. Strong process controllability
Fundamental Risks	Event nature alienation: controversy over "virtue not matching position", marginalizing professional judgment.	Selection tends to be conservative: may eliminate individuals with sharp personalities, leading to homogeneous results and lack of surprise.

6.2 Design Philosophy and Objectives: Balance, Integration, and Procedural Justice

An ideal mechanism should establish core competitive standards as the baseline for evaluation while ensuring **professionalism**. Through institutional design, it should effectively incorporate **public opinion** to enhance the event's vitality and influence. Ultimately, it must rely on a **clear and transparent** rule system to ensure all participants can intuitively understand and efficiently execute the process, thereby achieving a harmonious integration of professional depth, social breadth, and procedural clarity.

6.3 Proposal: Two-Stage Consensus Elimination Mechanism

This two-stage mechanism integrates **professional rigor** with **democratic participation** by first tasking a judging panel to nominate two contestants for elimination based strictly on technical performance—establishing a non-negotiable professional baseline—and subsequently empowering the audience to make the final decision between them, thereby institutionalizing public sentiment within a professionally validated framework to balance **competitive standards** with **collective influence**.

6.4 Mechanism Advantages and Expected Outcomes

This design achieves multiple benefits through **institutionalized checks and balances**: it not only effectively mitigates the public **perception risk** of "undeserved prestige," but also organically integrates the "**competitive threshold**" of professional judging with the "**entertainment appeal**" of public voting, transforming their opposition into complementarity. This mechanism infuses the elimination rounds with intense narrative tension and suspense. While enhancing the show's entertainment value, it ultimately builds a **dynamic consensus** that is both professionally persuasive and grounded in public sentiment, achieving logical coherence between elite judgment and popular choice.

7 Sensitivity Analysis

The model demonstrates strong robustness and reliability. As shown in Figure 13, even with high levels of random noise (up to 50%) added to votes, the predicted candidate rankings remain stable, with confidence intervals showing no significant overlap. Furthermore, Figure 14 indicates that the simulation results converge quickly, with about 80 trials being sufficient for stable outcomes. In short, the model effectively captures real competitive factors and produces trustworthy results for analysis.

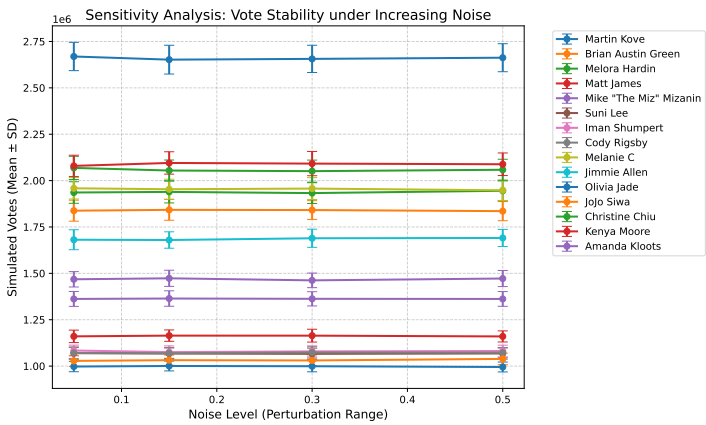


Figure 13: Vote Stability under Increasing Noise

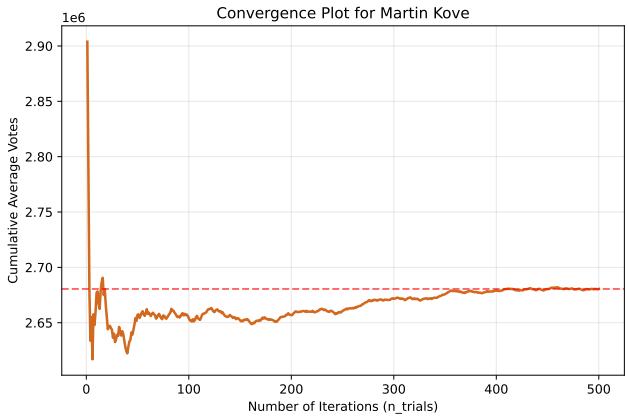


Figure 14: Convergence Plot for Martin Kove

8 Model Assessment: Strengths and Limitations

8.1 Key Strengths

The model offers the following advantages:

1. **Combines prediction and explanation:** It integrates the predictive power of machine learning with the explanatory capability of simulation modeling, enabling both outcome forecasting and uncertainty quantification.

2. **Provides systematic diagnosis and actionable improvements:** By comparing two voting mechanisms, it reveals their inherent differences and proposes an implementable two-stage voting scheme.

8.2 Limitations and Future Work

The main limitations of the model include:

1. **Data dependence and necessary simplifications:** Model performance relies on historical data quality, and complex social factors are simplified in the modeling

process.

2. **Limited generalizability:** Trained on data from a specific show, the model requires recalibration for application in different contexts or cultures.

9 Memo

To: Producers of DWTS

From: Team #2602045

Date: February 2, 2026

Dear Producers of DWTS,

I hope this letter finds you well. Preparations for the new season must be progressing at a busy pace—thank you all for your hard work.

I have noticed that the show has adjusted its competition format twice, and both changes seem to be related to the phenomenon of "highly popular contestants with relatively weaker technical skills advancing for extended periods." Whether under a percentage-based scoring system or a ranking-based one, it appears difficult to fully balance technical merit and popularity.

If the goal is to further strengthen the show's professional selection aspect, I would like to suggest considering a two-stage elimination system: first, the judges select two contestants for potential elimination based solely on technical performance; then, the audience votes to "save" one of them. This approach would allow professional judgment to serve as the primary basis for elimination while preserving a crucial role for audience participation, and it could also enhance the dramatic tension created by unexpected turns in the competition.

This design might offer a new balance between technical fairness and entertainment value. It is merely a small suggestion from a viewer, for your consideration during planning.

Wishing the show continued high ratings and ever greater success!

Sincerely yours,
Team #2602045

References

[1] U.S. Census Bureau. (2021). Intercensal estimates of the resident population for the United States, regions, states, and Puerto Rico: April 1, 2010 to July 1, 2020. https://www.census.gov/data/tables/time-series/demo/popest/intercensal-2010-2020-state.html

[2] Gershman, J. (2012). America's Idol? How the Contestant Most Voted for Doesn't Always Win. Quadrivium: A Journal of Multidisciplinary Scholarship, 4(1), Article 6.

[3] Garcia, M. T., & Tan, E. S. (2013). Regional Voting in a Philippine Television Song Competition. Philippine Sociological Review.

[4] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. In Advances in Neural Information Processing Systems 30 (pp. 4765–4774). Curran Associates, Inc.

Report on Use of AI

1. Google Gemini (Nov 18, 2025 version, Gemini-3)
   Query1: What is LightGBM?

   Output: LightGBM (Light Gradient Boosting Machine) is a distributed gradient boosting framework based on decision tree algorithms developed by Microsoft. By utilizing innovative histogram-based algorithms, a leaf-wise growth strategy with depth constraints, and techniques such as GOSS (Gradient-based One-Side Sampling) and EFB (Exclusive Feature Bundling), it significantly improves training speed and reduces memory consumption when handling massive datasets while maintaining high accuracy, making it one of the most mainstream and efficient modeling tools in current data mining and machine learning competitions.

2. Google Gemini (Nov 18, 2025 version, Gemini-3)
   Query2: How to use SHAP?

   Output: SHAP (SHapley Additive exPlanations) is a model interpretation method based on game-theoretic Shapley values, which quantifies and decomposes the process of moving a model from a "base mean" to a "final predicted value" by calculating the contribution of each feature to the prediction. The core advantage of SHAP lies in its ability to provide both global explanations (such as feature importance rankings) and local explanations for individual samples (showing how specific features push a prediction higher or lower), and because of its rigorous mathematical foundation, it can handle interactions between features.